The 8th Global Competition on Design for Future Education (2025)

Project Design Report

Project Name:

YOURTWIN—A Digital Clone for Cognitive Growth

Project Type：Solution Design

Competition Theme：
Growth Curriculum System Design for Primary Schools Towards 2050 (Holistic Human Development)

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Abstract（Chinese and English）

科技的快速发展导致了对现有技术的过度依赖，使得孩子们的创造力越来越少，思考能力越来越差。 目前，学生们正在依靠人工智能来为他们进行思考和决策。 预计随着更多技术的引入和发展，这种趋势可能会持续到 2050 年。 这会导致智力上的懒惰，当大脑不参与活动时，会存在认知退化的风险。 为了让学生为不确定的未来做好准备，需要强调实践技能、认知灵活性、元认知能力和批判性思维技能，这些技能在未来将具有价值。 YOURTWIN 是一种教育方法，它利用数字孪生、全息技术和量子计算来彻底改变小学学习。 该解决方案利用技术改造的房间，存储和处理数据，并投射 3D 互动全息图。 利用数字孪生技术，打造认知教育双胞胎，即镜像孪生和影子孪生。 镜像孪生是学习者画像的精确复制品，通过量子计算，作为众多学习方法模拟的主体。 另一方面，影子双胞胎被设计为学习者的全息对立面，学习者的弱点是它的优势，反之亦然。 该项目的使用工作流程包括数据收集、数字孪生生成、全息投影、认知挑战和反馈。 该项目的支持者将与多个机构、专家和社区合作，进一步支持拟议解决方案的开发。 随着技术以飞快的速度不断发展，开发所需的技术可能很快就会出现。 不过，在 2050 年之前，数字孪生、行为建模、全息技术和量子计算等技术取得进展之前，该项目仍是未来教育的解决方案设计。

The fast-paced evolution of technology has led to overreliance on existing technology, causing children to become less creative and less of a thinker. As of now, students are relying on artificial intelligence to do the thinking and decision making for them. It is projected that this trend may continue to 2050 as more technologies are introduced and become more advanced. This can lead to intellectual laziness and a risk of cognitive erosion when the brain does not engage in activities. To prepare students for an uncertain future, there is a need to emphasize practical skills, cognitive flexibility, metacognitive abilities, and critical thinking skills, skills that are going to be valuable in the future. YOURTWIN is an educational approach that utilizes digital twinning, holographic technology, and quantum computing to revolutionize primary school learning. This solution utilizes a technologically modified room that stores and processes data and projects 3D interactive holograms. Using digital twinning technology, two twins are created to foster cognitive education,

the Mirror Twin and the Shadow Twin. The Mirror Twin is an exact replica from the learner's profile, serving as the subject for numerous learning approach simulations through quantum computing. On the other hand, the Shadow Twin is designed as the holographic opposite of the learners wherein the learner's weaknesses are its strengths, and vice versa. The usage workflow of the project involves data collection, digital twin generation, holographic projection, cognitive challenges, and feedback. The proponents will partner with several agencies, experts, and communities to further support the development of the proposed solution. As technologies continue to evolve at a rapid pace, the technology necessary for development may be available soon. However, until the technological advancements in digital twinning, behavioral modelling, holographic technology, and quantum computing by 2050, the project remains as a solution design for the future of education.

1. Project Background

In the wake of the world's fast-paced evolution, technologies are becoming increasingly advanced. It is projected that by 2050, artificial intelligence (AI) would be so advanced that it will surpass the cognitive abilities of humans [1]. AI, alongside different technologies, evolved at a rapid pace where new technologies are introduced or becoming more advanced with each passing year. However, in the middle of these advancements, certain problems arise.

While the concept of machinery and technology revolve around the idea of making human lives easier, it has long been overused. Emerging technologies such as AI or automated systems unconditionally offer convenience and efficiency to human life. However, with each developed technology, humans tend to overuse and rely on them for convenience, leading to lowered critical thinking sessions and brain activities. In a survey, 86% of the student respondents are concerned about the lack of originality due to AI use, 75% are worried about AI limiting critical thinking skills, while 73% think about the possible overreliance on the technology [2]. This shows that the frequent trend of technology undermining a person's skills and abilities are observed not just by experts, but also by students who frequently use it. A study in 2024 showed that in foreign language learning, technology-reliant students showed lower language proficiency and less confidence in spontaneous language used [3]. A meta-analysis on the impact of technology highlights that continued use of smartphones and social media is correlated with poorer academic outcomes due to potential negative

effects on key cognitive skills for academic success [4]. Another closely related study concluded that relying on AI tools weakens one's critical cognitive skills such as decision making, critical thinking, and analytical reasoning [5]. Cognitive abilities are the ones most affected by overreliance on technologies. Due to limited brain activity when using these technologies, the brain is not trained to think for itself, relying on technology to decide.

As the world progresses even further, skills learned today will become obsolete in the future. Necessary skills needed in the future will change just as current jobs will also change. The World Economic Forum (WEF) 2023 noted that by 2030, problem-solving, critical thinking, adaptability, and lifelong learning would be the most valuable skills [6]. By 2050, these skills would change again while the world develops. With the rapid pace of development, the information and skills relevant today will hold no value in the future. The current education curriculum consists mostly of generalized education, a one-size-fits-all model, and a memorization approach teaching students how to be relevant today, but not in the future. Considering how AI and technologies, even today, cause cognitive erosion, there is a need to improve these aspects in a student's skillset. The best educational approach to prepare for the uncertain future is to create cognitive and emotional dispositions, and to improve one's skills to adapt [7]. What students need to learn is to develop the skills to continuously learn and think for themselves. For students to be future-ready, there is a need to develop cognitive adaptability and critical thinking skills starting from growth development from primary schools.

2. Design Process

In designing the project, the proponents used the design thinking methodology. In doing so, the project is planned systematically and is organized to ensure that it is well planned and well outlined.

2.1. Empathize

The design process began with understanding what skills are necessary in the future of 2050, and what problems should be prevented. In numerous articles, a global trend shows a problem in education wherein AI tools are continuously used by students. A UNESCO article notes that over two-thirds of secondary-school students in high-income countries regularly rely on AI tools to

complete academic tasks [8]. In the UK, 92% of students now regularly use generative AI tools, a sharp rise from 66% just a year earlier [9]. This trend is also locally observed by the proponents of the project wherein students would depend on ChatGPT to write essays and even solve trivial problems, while also copying answers from Google searches. Information technology and computer science students would also sneakily use AI to write lines of code in programming laboratories.

For the future of education, learners must prioritize attitudes, values, entrepreneurship, critical thinking, and metacognitive skills to thrive in a changing world, according to OECD [10]. According to the recommendations of these articles, there is a need to alleviate the effects of AI and technologies, particularly its effects on one's cognitive abilities.

2.2. Define

Following the empathetic findings, by 2050, the problems of overdependence on technologies are projected to continue, thus promoting further cognitive erosion. As computers become smarter, there will come a time when children might not know how to think for themselves, solve new problems creatively, or make tough decisions on their own. This trend is expected to continue as technology becomes more sophisticated and evolves over time. And with the constant changes in the world, relevant skills and ideas right now may become obsolete in the future. For children to become future-ready, what's important is not mastering the skills presented today, but the skill to not stop and keep learning. Children need to learn how to keep learning, adapt, and even unlearn old skills to be ready for the future. Education systems will need to prioritize practical skills, problem-solving, and emotional intelligence over just memorization to prepare students for an evolving environment.

2.3. Ideate

After the definition of problems, the proponents began structured brainstorming sessions to build a viable solution by 2050. The ideation phase was guided by the concept that the proposed solution must not only alleviate cognitive erosion and overreliance on technology, but also improve students' capacity for adaptability, creativity, and independent thought. The solution must maintain focus on the mitigation of cognitive erosion and overdependence on technology while enhancing adaptability for students to be future-ready. Following the suggestions of experts from mentoring sessions, the idea must adhere to the following criteria.

a. Alignment to the Cognitive Flexibility Theory - ensuring that the solution promotes learners the ability to view and learn new content from multiple perspectives and approaches.
b. Integration of Metacognition – ensuring that the solution encourages students to question and think about their learning process to stimulate critical thinking.
c. Feasibility in a 2050 Educational Context – designing the solution with consideration for projected technological capabilities and learning environments by 2050.

Through this ideation process, the proponents decided on a concept that combines advanced technology with human-centered learning, ensuring that future learners are equipped not only with knowledge but also with the cognitive skills to thrive in a rapidly changing world.

2.4. Prototype

Building from the polished idea from the ideation phase, the developed solution design is named "YOURTWIN". The proposed concept demonstrates utilizing digital twinning technology to create personalized learning experiences for students. The conceptual design includes the implementation of emerging technologies that are projected to be improved and more advanced by 2050, including digital twining, advanced behavioral modelling, holographic technology, and quantum computing capabilities. While the current stage focuses on conceptualization rather than physical implementation, the prototype framework is detailed to illustrate its functional components, technological underpinnings, and potential user interactions. Further technical and structural details of YOURTWIN are provided in Section 3. Introduction to Works.

2.5. Test

The testing phase will be designed to evaluate YOURTWIN's effectiveness in enhancing cognitive adaptability and independent decision-making. This phase would be conducted once the technologies required to develop the foundation of the solution are more capable and more advanced. The product solution would be pilot tested to 3 primary school classrooms first before further advancements and deployment. The target participants would be primary school students currently in their growth development cycle. The planned approach would be focusing on both technical validation and educational impact assessment. It will follow the following criteria.

Table 1. YOURTWIN Assessment Criteria

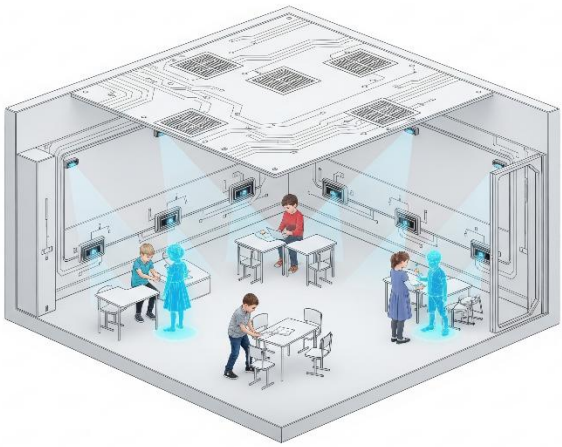
Criteria	**Description**	**Measurement**
Cognitive Adaptability	Ability to adjust thinking and learning method when facing new problems	Pre- and post-assessment using cognitive flexibility tests
Critical Thinking	Capacity to analyze and create solutions from different perspectives	Rubric-based evaluation of student responses to challenges
Twin Accuracy	How closely the twins match the learner's real profile	Comparison of twin behavior using base student data as reference, and expert reviews
Engagement and Motivation	Level of interest and active participation in lessons	Observation logs and self-reported motivation surveys
Learning Effectiveness	Improvement in knowledge retention and understanding of concepts	Standardized test scores and practical application
Simulation Response Time	Speed and accuracy of the quantum computing system in generating optimal learning strategies	System performance logs
Collaboration with Shadow Twin	How well the learner interacts and learns with the shadow twin	Peer-like interaction quality ratings and analysis of peer-to-peer learning outcomes
User Experience	Overall satisfaction with the YOURTWIN environment	Surveys, interviews, and usability testing feedback

Following the testing stage, it is expected that it will provide verification that the system's algorithms adjust effectively to individual learners, evidence of enhanced cognitive capabilities, and usability insights for further refinement and maintenance. This testing approach ensures that once

developed, YOURTWIN will be tested to further improve its design and educational value.

3. Introduction to Works

YOURTWIN is a solution design for teaching cognitive adaptability and critical thinking skills to primary learners as a part of the growth curriculum of primary schools. It is an educational approach that fully utilizes the digital twinning technology to revolutionize the learning experience of primary school learners. Through the advancements of technology by 2050, a new form of teaching and learning will be formed. YOURTWIN utilizes a whole technologically modified room to conduct holographic learning experiences, wherein the walls and ceiling of the room have a computer that can store data and can project 3d interactive holograms. This modified room will actualize a digital classroom and will bring the learner's digital twin to life. The objective of this approach is to develop the cognitive adaptability and critical thinking of primary school learners to be more future-ready.



Picture 1. Concept art of YOURTWIN

3.1 Features and Functionalities

In building and developing YOURTWIN, the following features will be implemented alongside various functionalities that serve different functions for a cohesive and adaptive learning system.

a. Digital Twin Module

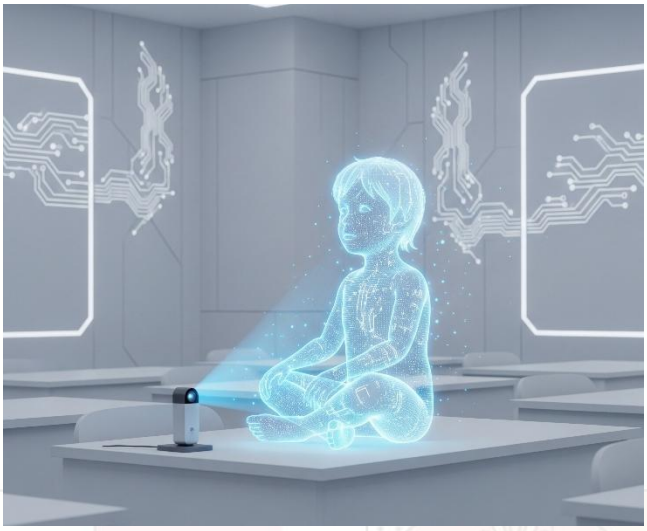
This core functionality utilizes advanced biometric, behavioral, and emotional data to create two distinct replicas of each learner, the Mirror Twin and the Shadow Twin. The Mirror Twin is an exact clone of a learner's personality traits, behavioral patterns, and

consciousness, serving as a subject for learning simulations to determine the best learning modality for the student. This twin remains inside the computer as pure data only for learning simulations. The Shadow Twin is designed as the cognitive opposite of the learner, wherein the learner's weaknesses are the Shadow Twin's strengths, and the learner's strengths are the Shadow Twin's weaknesses. This twin is the one that is projected through the holographic system as a peer classmate of the learner. The objective is for the learner and the Shadow Twin to conduct peer-to-peer learning and learn from each other for holistic development.

Picture 2. The Shadow Twin learning with the student.

b. Immersive Holographic Projection System

Within wall and ceiling panels, micro-projectors and nano-cameras are embedded and integrated discreetly. These devices create the 3D life-sized holographic Shadow Twin in real-time, allowing learners to physically interact with their Shadow Twin in a shared space. The hologram it provides is dynamic and could be interacted with alike with human interactions. The minimalistic design of the panels ensures that the room appears free of visible technology while housing an advanced computing ecosystem behind the surface.



Picture 3. Conceptualized image of a hologram twin.

c. Adaptive Learning Simulation Engine

Through quantum computing and real-time data processing, this engine runs thousands of learning simulations using the Mirror Twin to determine the most effective teaching strategies and learning approaches during active sessions. The system continuously adjusts challenges and interaction styles based on the learner's current performance and cognitive state. The data is then reflected as to how the Shadow Twin teaches and interacts with the learner to ensure that the student is learning with the best approach possible.

d. Cognitive Challenge Modules

A library of interactive learning activities is integrated within the Shadow Twin's behavior patterns. These modules give the Shadow Twin the ability to think differently from the learner. Thus, it forces the learner to engage in alternative reasoning and counterarguments. By using unconventional problem-solving approaches, it reinforces the student's metacognitive skills and cognitive flexibility.

e. Privacy and Ethical Safeguards

Given the deep personal nature of the data required for YOURTWIN, it is understandable that confidentiality is a concern. Therefore, YOURTWIN will incorporate secure, encrypted data handling protocols, wherein all personal information remains within the system unless authorized for research purposes. The models will be designed with accountability features to prevent abuse and misuse within the system.

f. Minimalist, Children-Friendly Environment

The classroom environment is designed to reduce cognitive overload by maintaining a clean space with minimal distractions. All technological elements are integrated seamlessly into the walls and ceilings of the room with no visible wirings, large screens, or machinery visible. Ambient lighting will then further optimize the learner's focus and engagement on lessons.

3.2 Functional Workflow Overview

YOURTWIN is designed to be an adaptive and immersive learning environment that leverages advanced technology to deliver personalized cognitive education to primary learners. The following outlines the step-by-step functional workflow of the system from student arrival to progress tracking.

1. Student Arrival

Upon entering the YOURTWIN classroom, each learner undergoes a secure data capturing process. This captures biometric data, behavioral patterns, and emotional data. The data collection process is designed to be quick and comfortable and ensures student privacy through encryption protocols. Assuming that these data would be available publicly in 2050, the system then proceeds to the next phase.

2. Digital Twin Creation

The collected data is processed by the Digital Twin Module, which then generates two digital twins per learner, the Mirror Twin and the Shadow Twin. The twin generation happens simultaneously for all students present for their own personalized learning session.

3. Holographic Projection and Interaction

The classroom's micro-projectors and nano-cameras activate to project the 3D hologram of each learner's Shadow Twin. The space between the holograms are managed to allow multiple students to interact with their own Shadow Twin simultaneously. Learners can physically interact, converse, and collaborate with their holographic classmates in a natural way.

4. Adaptive Learning Simulation

YOURTWIN's simulation engine uses the Mirror Twins to conduct thousands of learning simulations in real-time. These simulations assess each learner's cognitive state and

determine the best teaching strategies and learning approaches. Based on the results of the analysis, the Shadow Twins adjust their behavior, challenges, and interaction style to maximize the cognitive development of learners.

5. Cognitive Challenge Engagement

   The modules within the Shadow Twins are designed to promote metacognition and flexible problem solving. Because Shadow Twins are the opposite of the learners, they encourage the exploration of new perspectives and improve critical thinking through peer-to-peer learning and cooperative tasks.

6. Monitoring and Progress Tracking

   YOURTWIN continuously monitors learners' performance during sessions. Immediate feedback is provided via the holographic Shadow Twin, promoting self-awareness and motivation. Educators can access the summarized progress reports to track the growth of each learner over time.

4. Feasibility Analysis

The feasibility of YOURTWIN depends on the integration of multiple emerging technologies, each contributing to a cohesive, future-ready learning solution. This section reviews developments and analyzes the technical, market, risks, and impact feasibility. While some of the features are still in the early stages of adoption, their research paths show that YOURTWIN's core functions can be implemented by 2050. In order to ensure readiness, scalability and adaptability, in an evolving educational landscape, YOURTWIN leverages both current capabilities and anticipated technologies.

4.1 Technical Analysis

Digital Twinning

YOURTWIN is based on the concept of digital twins to create a personalized learning experience for each child using the Mirror Twin and the Shadow Twin. Saeed et al. [11] explored what our cities of the future might look like, focusing on how digital twins real-time, virtual replicas of urban environments could transform the way cities are planned and managed. Unlike static maps or models, digital twins can constantly update through data from sensors, infrastructure, and even

everyday city users, who act as “mobile sensors” by sharing information through their interactions and movements. The authors envision future cities as more connected, sustainable, and responsive, with systems that can predict issues, improve services, and involve citizens directly in decision-making. While the potential is huge, they note that real-world, large-scale applications are still in their early stages, leaving plenty of room for innovation and research [11]. This means that with he advancements, YOURTWIN would be able to adjust lessons, challenges, and feedback in real time based on the child’s progress and engagement. YOURTWIN will apply the same principle of prediction to predict learning gaps and maintain motivation for students.

As 6G networks become more advanced, managing them is getting harder because there are so many different devices, services, and user needs. Traditional management methods can’t keep up with the demands for speed, low delay, and reliability. To solve this, the authors propose a new system called Intent-Based Network Digital Twin (IBN-DT). It uses a virtual copy of the network to test and manage changes before applying them in real life, making the process more accurate and efficient. The framework separates decision-making (knowledge) from direct control, processes requests at multiple levels, and runs automatically without constant human input. In testing, it increased revenue by 40% in situations with limited resources [12].

Behavioral Modelling

Understanding how students behave and make decisions is important and significant to cognitive adaptability. Stern and Richardson [13] review research on how road users such as drivers, cyclists, and pedestrians behave, highlighting the importance of understanding human decision-making for improving traffic safety and efficiency. They discuss existing behavioral models, point out gaps in knowledge, and outline future research needs, such as incorporating psychological factors, real-time decision-making, and interactions between different types of road users. Their work emphasizes that better behavioral models can lead to more effective transportation planning and policymaking [13]. YOURTWIN will use behavioral modelling to figure out how a learner solves problems, how they feel about things, and how they like to learn. This lets the twin be a

responsive learning partner who can give the other twin personalized challenges that help them think critically.

Holographic Technology

Shvetsov and Alsamhi [14] explore how holographic communication, a technology that creates lifelike 3D interactive experiences, can be integrated into the Metaverse to make virtual interactions more realistic and engaging. They describe potential applications across industries, including education (virtual classrooms and 3D simulations), business (immersive meetings and product demos), entertainment (gaming and 3D broadcasting), healthcare (remote consultations and surgical training), and remote assistance (real-time holographic support). The paper also examines challenges such as the need for better AI-driven content optimization, advanced compression methods, and emerging technologies like terahertz communication and quantum holography. The authors highlight that merging holography with Metaverse could significantly transform digital engagement in the near future [14]. In YOURTWIN, the holographic twin would look exactly like its real version, except for a distinguishable colored outline to identify the learner and twin. It could also be placed in a natural way in the learner's physical space to make it possible to do demonstrations in real time and take part in group tasks.

Huang and Cao [15] review recent progress in quantitative phase imaging (QPI) using holography, a technique that captures detailed information about a sample's structure without the need for labels or dyes. They discuss how advances in holographic methods have improved imaging resolution, speed, and data analysis, making QPI more powerful for studying biological cells, materials, and microstructures. The authors also highlight new trends such as integrating AI for image reconstruction, combining QPI with other imaging modalities, and developing compact,

portable systems for real-world applications [15]. In practice, this means that a student could ask the twin a math question, watch it draw the answer step by step in the air, and get personalized feedback, all in the same interactive session.

Quantum Computing

Möller and Vuik [16] examine how quantum computing could shape the future of high-performance scientific computing. They discuss the fundamental differences between quantum and classical computing, noting quantum systems' potential to solve certain problems—such as complex simulations and large-scale optimizations—much faster than traditional methods. The authors also highlight ethical and practical considerations, including the challenges of developing quantum algorithms, the risk of creating technological gaps between nations, and the need for interdisciplinary collaboration to ensure responsible use. Overall, the paper presents quantum computing as a promising but still maturing technology that could significantly expand the capabilities of scientific research in the coming decades [16]. This processing power would also help with predictive modeling. The system could predict how the skills of learners would develop over time, find potential learning problems before they happen, and change how it teaches based on what it learns from the user data. For example, if the model sees that a student's logical reasoning skills aren't getting any better, it could change the curriculum right away to include more critical thinking challenges. It could also keep the student interested by using personalized storytelling or game mechanics.

Table 2. Summary of Technical Feasibility

Technology	**Relevant Study**	**Feasible**
Digital Twinning	Saeed et al. [11], IBN-DT [12]	✓
Behavioral Modelling	Stern & Richardson [13]	✓
Holographic Technology	Shvetsov & Alsamhi [14], Huang & Cao [15]	✓
Quantum Computing	Möller & Vuik [16]	✓

4.2 Market Analysis

The global education technology market is expanding rapidly, driven by demand for more engaging and personalized learning tools. Organizations like the World Economic Forum [6], OECD [17], and UNESCO [18] stress the importance of critical thinking, problem-solving, and adaptability in future jobs of all skills YOURTWIN aims to develop. By 2050, schools, governments, and private institutions will actively seek adaptive tools like YOURTWIN to prepare students for an AI-driven world. The system could be marketed to public schools, private academies, and even corporate learning programs, making it viable across multiple sectors.

4.3 Risk Analysis

YOURTWIN has a lot of potential, but there are challenges and several risks that can be thought of. And these are the following:

1. Technical risks

   Dependence on maturing technologies like advanced holograms and quantum computing may delay rollout. Holographic projection is impressive as the moment, but still expensive right now, but in the year 2050, it will likely be more advanced and improved, somehow if

the cost remains high, it will delay the implementation. There is also quantum computing, well it is in its early stages. Companies like Google and IBM are creating breakthroughs for widespread commercial availability. If ever quantum computing lags into adoption, the system will need an alternative computing solutions but still could reduce its intended performance. Therefore, delays of this technology will cause the system to push back the project timeline.

2. Financial risks

High initial investment for equipment, infrastructure, and skilled staff could be challenging for some schools. This includes specialized, or high-end projectors, sensors, high-performance computing systems, and secure data storage facilities If funding from the government or of private investors is insufficient, some schools may not be able to have an operation of the system to launch.

3. Data privacy risks

Collecting biometric and emotional data requires strict compliance with privacy laws [19]. Noncompliance to this could lead to lawsuits, heavy fines, and lost of public trust. Each school will need to imply robust cybersecurity systems to protect the data from any unauthorized access or any form of hacking. Because this kind of data is sensitive and, if leaked, it could really cause serious harm to the student's privacy. Data Security standards must meet these standards in every location to ensure the safety of each student's data and information.

4. Operational risks

Teachers and students need training to use the system effectively. Faculties must learn

first how the system works and remodifies their lesson plans to integrate the holographic and AI-driven mode of teaching. While students should experience the system to adapt into the technology; first try might feel unusual. This is because lack of readiness can lower the systems effectiveness as the teachers might underuse it and the students might not fully benefit from YOURTWIN. This will lead to wasted investment in technology if it is not used in its full potential.

5. Adaptation risks

Some institutions may resist major changes to traditional teaching [20]. Some of the institutions can be culturally attached to face-to-face (F2F) classes with teachers facilitating them; they might view that the system is a replacement of human teachers rather than enhancing only the education and learning experience. Due to this, legislators and education boards may also hesitant and slow to approve such a disruptive system for official use. Other factor are the parents, who may be concerned about too much reliance on technology in the education system. To sum up, even the system YOURTWIN can work perfectly, slow in adoption and adaptation could delay the larger scale impact of the system, limiting its market penetration in the early phase.

These risks can be reduced through phased implementation, partnerships with tech companies, strong privacy measures, and integration with traditional methods. If these implementations are completely implemented and the adoption is fast and duly recognized. The system will be a widely used technology that all can benefit from the future-goal technology.

4.4 Impact Analysis

YOURTWIN can change the way kids learn by making lessons more personal, hands-on, and flexible for each student. Instead of everyone learning the same way, YOURTWIN adjusts to a

student's own pace, interests, and strengths. This helps them think for themselves, find creative solutions, and keep up with fast changes in the world—skills experts say are needed for future jobs [6][17]. Studies show that when lessons are personalized, students can be up to 30% more engaged and improve their grades by 21% [21]. The hologram technology in YOURTWIN also makes learning more exciting—research found students are 3.8 times more emotionally involved and remember 75% of what they learn, compared to only 5–10% in normal lectures [22].

Aside from helping students learn better, YOURTWIN could push new inventions in education technology. Its mix of AI, holograms, and smart learning tools could inspire more advanced teaching methods, just like how introducing tablets in schools changed the way classes are taught worldwide [18]. If made affordable for everyone, YOURTWIN could also close the gap between rich and poor students by giving all children the same chance to get high-quality lessons. This supports the global goal of making education fair and equal for all students, no matter where they come from [17][18].

5. Innovation

YOURTWIN revolutionizes the education landscape by incorporating emerging technologies for students to be more future-ready by 2050. These innovations are important to ensure that the educational curriculum does not remain static. By implementing several innovations, YOURTWIN is able to transform how students learn and develop essential future-ready skills. From Digital Twinning to Adaptive Learning, the innovations introduced by YOURTWIN are very groundbreaking in transforming primary school learning.

5.1 Personalized Learning with Digital Twin

Unlike traditional teaching and learning methods that often apply a one-size-fits-all approach, YOURTWIN creates personalized twin for every student. Learners understand these twins more as their personality and behavior are just derived from theirs. On the other hand, these twins understand the learner's unique strengths, weaknesses, and behaviors. This allows learning to be customized dynamically, ensuring that every student receives the best teaching style and learning approach for them. This is allowed by generating two digital twins for each student, the Mirror Twin and the Shadow Twin. The Mirror Twin serves as an internal data model that is used exclusively for running learning simulations to identify the most effective learning methods for the student's cognitive needs. On the other hand, Shadow Twin acts as an interactive, holographic peer that is designed to

challenge the learner to think from different perspectives, promoting holistic growth. This approach advances personalized learning beyond adaptive software by integrating continuous simulation, emotional intelligence, and real-time adaptation.

5.2 Immersive Holographic Interactions

An innovation of YOURTWIN is its fully immersive 3D holographic projection system that brings learning life inside a technologically modified classroom. Through the embedded micro-projectors and nano-cameras within the walls and ceilings, YOURTWIN projects interactive holograms of each student's Shadow Twin into the room. These holograms act as peer classmates who learners can interact with through natural gestures, speech and shared activities. This goes beyond traditional virtual reality (VR) or augmented reality (AR), which often require bulky equipment and headsets, by integrating holograms into the student's real environment without intrusive devices. This results in a blend of physical and digital worlds, creating a learning environment that enhances engagement and motivation. This can also develop social and communication skills due to naturalistic interactions with digital peers, making learning both effective for cognition and emotion.

5.3 Real-Time Adaptive Learning

YOURTWIN introduces an advancement in adaptive education through its quantum-power, simulation engine. Through quantum computing, the system runs thousands of learning simulations all at once using the Mirror Twin's model. These simulations test and evaluate multiple teaching and learning strategies and interaction modalities to determent the best approach for each learner. As a result of the said simulations, the Shadow Twin's behavior, teaching style, and interactivity adjust in real-time according to the student's performance and cognitive load. This continuous learning and adaptation maintain that the learner is challenged enough to stimulate growth but not too difficult as to cause frustration or disengagement. This innovation surpasses linear teaching models by providing a responsive educational experience that adapts to the learner's needs, promoting cognitive development and critical thinking skills necessary for the future.

5.4 Cognitive Flexibility through Peer Learning

Another key innovation in YOURTWIN is the concept of oppositional peer learning, conducted by the Shadow Twin. Unlike traditional tutoring systems, the Shadow Twin is designed as the

student’s opposite, where the student’s weaknesses become the twin’s strengths, and the student's strengths become the twin's weaknesses. This contrast introduces conflict and diverse viewpoints into the learning process, encouraging learners to reconsider their solutions and expand more on their problem-solving approaches. By engaging with a digital peer that challenges their thinking patterns, learners become more aware of their own thinking process. This develops cognitive flexibility, allowing students to adapt more effectively. This challenge stimulates critical thinking and creativity in ways that traditional education cannot achieve, preparing learners for the uncertainties of the future world.

6. Summary and Plans

This section provides a concise summary of YOURTWIN and its core components. It also offers a balanced analysis of the project’s strengths and weaknesses. Finally, it outlines future directions and development plans in order to further advance YOURTWIN toward deployment and implementation and to expand its educational impact.

6.1 Project Summary

YOURTWIN is a solution designed to teach cognitive adaptability and critical thinking skills to primary learners as part of the growth curriculum of primary schools. This educational approach utilizes digital twinning technology to revolutionize the learning experience of primary school learners. YOURTWIN utilizes a technologically modified room to conduct holographic learning experiences, where the walls and ceiling have a computer that can store data and project 3D interactive holograms. The objective is to develop the cognitive adaptability and critical thinking of primary school learners to be more future-ready. The features of the system include a digital twin module that uses advanced data to create the Mirror Twin and the Shadow Twin. The Mirror Twin is an exact clone of a learner’s traits, while the Shadow Twin is designed as the opposite of the learner. While the Mirror Twin serves as a subject for quantum learning simulations, the Shadow Twin is projected as a holographic peer and learning companion of the student. Cognitive challenge modules within the Shadow Twins also encourage learners to engage in alternative reasoning and counterarguments, reinforcing their metacognition skills and cognitive flexibility. This aligns with the Cognitive Flexibility Theory and Metacognition Theory, which test learners' ability to approach

complex situations from different perspectives.

6.2 Advantages and Disadvantages

With the innovations within the system, YOURTWIN offers numerous benefits that significantly enhance the learning experience, but it also presents certain challenges and limitations. Understanding the strengths and weaknesses of the project is crucial for evaluating the system's overall impact to guide its future development. The following are the advantages of YOURTWIN.

a. Personalized Learning Experience

YOURTWIN's dual digital twin system creates a personalized educational approach by continuously adapting to each learner's cognitive and behavioral data. Unlike traditional education that utilizes a one-size-fits-all model, YOURTWIN customizes learning by personalizing content, difficulty levels, and learning approaches that are optimal to each student's learning profile. This customization promotes better immersion and improved learning outcomes in comparison to traditional education. As a result, students will experience improved retention, motivation, and overall academic performance.

b. Immersive and Interactive Learning Environment

The implementation of interactive holograms within the classroom transforms learning and education into an immersive and engaging learning experience. By projecting interactive Shadow Twins, YOURTWIN creates opportunities for learners to engage in interactions with their digital counterparts. This environment encourages social interactions, enhances communication skills, and better comprehension of the learning materials.

c. Dynamic and Real-Time Adaptation

Through quantum computing, YOURTWIN dynamically adjusts teaching and learning approaches and interaction tyles in real time. This keeps learners in a zone where they are constantly challenged and supported. This helps prevent boredom and frustration, encouraging continuous cognitive growth.

d. Development of Cognitive Flexibility and Critical Thinking

Through interaction with the Shadow Twin, students develop cognitive flexibility and critical thinking. This opposition encourages students to solve problems from different perspectives and explore different problem-solving methods. Through this process, learners

reflect on their own thinking methods and adapt them to uncertain situations, developing metacognitive awareness. These skills are essential for problem solving and adapting to uncertain future environments.

e. Multi-User Capabilities

YOURTWIN is designed to accommodate multiple learners within a single classroom. Each student is paired with their own personalized Shadow Twin hologram. This allows the students to engage in a dynamic and interactive group setting environment. By supporting multiple holograms, YOURTWIN promotes collaborative learning where students can conduct group discussions and cooperative problem-solving along with their digital twins.

In addition to these advantages, YOURTWIN may also face some challenges in development and implementation. The following are the disadvantages of YOURTWIN.

a. High Initial Cost and Technological Complexity

The implementation of YOURTWIN requires advanced hardware such as quantum computing technology, micro-projectors for holography, and advanced biometric sensors. In addition to that, complex software systems are also required for real-time simulation and interaction. The combination of these technologies results in a high cost which needs a significant financial investment that may be prohibitive in schools. However, it is expected that by 2050, the technological complexities of YOURTWIN will be solved.

b. Technical Maintenance and Support Requirements

Maintaining a fully operational YOURTWIN classroom involves continuous technical monitoring, including updates and calibration. Due to the advanced technologies used such as holographic projection and adaptive learning engines, specialized technicians must be present to provide timely support. This kind of maintenance increases the operational costs and may present challenges for schools that lack adequate infrastructure, staff, and funds.

c. Data Privacy and Security Risks

Although YOURTWIN implements strong security measures, handling personal data still introduces risks. Data breaches, cyberattack, or unauthorized data sharing could lead to misuse of personal information. This could cause potentially serious ethical, psychological, and legal consequences for learners. This risk of harm increases if proper security measures

are not continuously maintained. Thus, continuous monitoring and compliance with data protection guidelines are necessary to mitigate these risks.

d. Potential Overreliance on Technology

While YOURTWIN offers powerful digital tools for personalized learning, it is still a concern with excessive dependence on technology. Overuse might reduce teacher-student interactions and may limit opportunities for students to learn social-emotional skills through real human interactions. It is necessary to balance the use of technology with the traditional humane methods to maintain a holistic education experience for learners.

e. Learning Curve for Educators and Learners

The innovations within YOURTWIN mean that teachers, students, and school administrators are required to invest time and effort to become proficient in operating the system. Teachers will need training to understand how to fully use the system as a leverage for personalized teaching and how to include the system in the existing curricula. In similar to that, learners are required to attend orientations to engage fully with the new learning modality. This learning curve may delay full implementation and may also consume institutional capacity and budgets.

6.3 Future Plans

As of now, the technologies required to fully develop YOURTWIN are not yet mature enough. There is still a need to advance some key technologies such as digital twinning, holographic technology and quantum computing. Thus, the proponents plan to be involved in the development of these technologies by providing active collaboration, continuous research and strategic partnerships with experts and institutions specializing in these fields. The idea will be pitched to potential partners, universities, research centers, and technology companies to convince them to participate in ongoing studies and projects related to the necessary technologies. Through contributions and collaborations, insights and findings would be shared to scientific and education communities through publications and conferences to further accelerate the pace of innovation and gather valuable feedback.

When the necessary technologies are finally advanced enough and available for use, the organized committee will then start the development of simplified versions and the key components

of the system. These prototypes will then serve as models that would guide full-scale development. During which in this time, the proponents would actively establish partnerships with the government, education sector, experts, and stakeholders for backing and support. When there is enough funding and backing, the proponents and the organized committee will engage in the full-scale development, testing, and deployment of YOURTWIN. This will involve active collaborations with developers to integrate mature digital twinning, holographic projection, and quantum computing systems into the educational framework.

In the time YOURTWIN is ready for deployment, the system will be tested according to the criteria given in Section 2.5 Test. At this time, the committee will still be involved in fine-tuning the system and pilot programs to evaluate the system's reliability and for maintenance. Intensive research and data collection would still be conducted to continuously improve YOURTWIN. Educators and administrators of primary schools would be first oriented and trained to equip the school with the knowledge and skills necessary to maximize the educational benefits of YOURTWIN.

7. Team Profile

MARC JOSEPH B. SACOPASO

Team Captain, Head Project Manager

Mariano Marcos State University
College of Computing and Information Sciences
Bachelor of Science in Information Technology

Marc Joseph is a very outstanding and intelligent individual. He is the consistent valedictorian of his class, making him more than capable enough to be the captain of Team SEEDED. Through his valiant leadership, Team SEEDED overcame a lot of obstacles and challenges. He is a very

skilled coder, and a very cool-headed person that can make a way in every type of adversity. He values his teammates very well, just as his teammates are grateful for his leadership.

KURT ANGELO D. AGCAOILI

Statistics Analyst, Foreign Affairs Coordinator

Mariano Marcos State University
College of Computing and Information Sciences
Bachelor of Science in Information Technology

Kurt Angelo has a unique charisma that makes communication between two people so light, easy, and understandable. If Marc Joseph was the brain of the team, Kurt Angelo is the mouth and ears of the team. His ability to win the good side of the people makes him stand out in his own way. Because of him, SEEDED was able to effectively communicate with different people, making him an important asset to this team.

ANGELO LAURENCE A. PALTING

Lead Designer, Software Architect

Mariano Marcos State University
College of Computing and Information Sciences
Bachelor of Science in Information Technology

Angelo Laurence is very talented when it comes to digital art and has the skill in the way that

he can see through design and aesthetics like it was dancing on the palm of his hands. With his skills, Team SEEDED produced magnificent visual outputs, impactful presentations, and futuristic art styles. Not only that but he is very skillful at coding and videography, he really holds the title of a MMSU-CCIS InfoTech Vice President because the capabilities he can show are top-notch.

MELJAN CHRISTOFFER M. CRISOSTOMO

Financial Manager, Feasibility Analyst

Mariano Marcos State University

College of Computing and Information Sciences

Bachelor of Science in Information Technology

Meljan Christoffer has a unique ability to make presentations and ideas come to life when it comes to other peers. With his concise explanations and reports, Meljan Christoffer can ace any type of presentation without losing composure. With his bright ideas, he also acts as the second brain of the team. Add it up with his fast-calculating skills and critical thinking, he is an asset to be reckoned with.

8. Appendix

Appendix 1. Analysis Report on AIGC Applications in Educational Project Design

Gemini Pro

As an all-rounder AI, Gemini Pro was used extensively in almost all aspects of the design process of YOURTWIN. It was mostly used to gather information and insights for the project. In addition to that, the integration of Google search into Gemini makes it finding references, especially websites, more efficient. Using Deep Research functionality, it could even find credible and significant articles for research papers. Gemini Pro was used continuously to ensure that everything we write and include in our project is accurate and feasible.

Sample Prompt: *"List down psychological theories that supports a claim that learning with a complete opposite of you will improve your cognitive adaptability. How does it improve critical thinking skills when learning with a complete opposite of you? Include psychological theories to prove it."*

ChatGPT

ChatGPT is another all-rounder AI that is quite popular on the global scale. It served as our backup AI generating tool in case Gemini Pro is not available. In times when Gemini Pro creates incomprehensible answers, we use ChatGPT to omit useless facts and explain only the necessary things. ChatGPT also helped in improving our paper by acting like a paper critic and suggesting things on how to polish our paper more. We have also used ChatGPT occasionally to summarize things when Gemini is not working.

Sample Prompt: *"What graphs can I use to minimally show the ratings on Educational Impact, Scalability, Community and Sustainability"*

Imagen

Imagen is a text-to-image model used by Gemini for image generation. It is our primary tool for image generation in the project. With it, nonexistent technologies come to life as it illustrates even the future. We have used it periodically whenever we needed to create a figure, a design, or an image.

Sample Prompt: *"Generate an image of a futuristic white-background blueprint with fine silver-gray linework, showing a spacious high-tech classroom in isometric cutaway view. The walls are covered with large rectangular acoustic noise-reduction panels, each drawn in detail to reveal that they are fully integrated computers."*

Veo

Veo is a text-to-video model also used by Gemini for video generation. It is the only video-generating AIGC we have used throughout the competition. Through Veo, we have generated numerous scenes and edited them together to form our video presentation. We only occasionally used it when necessary and were not used in things other than the video presentation.

Sample Prompt: *"Show a fast-paced montage of children using gadgets passively, scrolling, watching, staring blankly at screens. Blend in glitch effects or static to show overreliance and fading creativity. Narration: 'In a world evolving faster than ever, children are slowly forgetting how to learn… relying more on tech, and less on thinking.'"*

Canva AI

Canva AI is another image generation AI that we used to create intricate designs and prototypes for our project. The usage is very similar to Imagen, with the tool creating images that fully showcases the future. Since we already used Imagen for our project, we have rarely used Canva AI, only prompting us to use it whenever we need an instant image for our presentation.

Sample Prompt: *"Generate an image of a futuristic classroom with holographic digital twin assistant teaching children, soft lighting, vibrant colors, high detail."*

Table 3. Phases of Educational Project Design

Phase	**Design Scenario**	**AIGC Tool Used**
Topic Selection Phase	1. Topic Research	N/A
	2. Requirement Analysis	Gemini Pro, ChatGPT
Design Phase 1: Creative Ideation	3. Brainstorming	N/A
	4. Solution Decision-making	Gemini Pro, ChatGPT
Design Phase 2: Prototyping	5. Prototyping (Physical Model, Hardware Sketch, Software Prototype)	Imagen, Canva AI
	6. Work Evaluation	Grammarly, ChatGPT
Expression Phase	7. Design Report Drafting	Gemini Pro, ChatGPT
	8. Visual Presentation (PPT, Posters, Videos)	Imagen, Veo, Canva AI

Appendix 2. Supplementary Materials

The following supplementary files and materials are included in the attached folder.

1. IMAGE 1 - YOURTWIN CONCEPT.png – an AI-generated image that showcases the room

concept of YOURTWIN.

2. IMAGE 2 - CONCEPT ART.png – an AI-generated concept art that showcases learning with a Shadow Twin.
3. LOGO (Main).png – the main logo of YOURTWIN to symbolize the Mirror Twin.
4. LOGO (Secondary).png – the secondary logo of YOURTWIN to symbolize the Shadow Twin.
5. POSTER.jpg – a simple and concise poster to tease YOURTWIN.
6. PRESENTATION (PDF Version).pdf – the PDF version of the YOURTWIN presentation for reading and viewing.
7. PRESENTATION.pptx – the PPTX version of the YOURTWIN presentation for pitching and presentations.
8. PROJECT INTRODUCTION and TEAM INFORMATION.docx – a DOCX document containing the project introduction of YOURTWIN and the team information of SEEDED.
9. VIDEO.mp4 – a 1-minute video showcasing the concept of YOURTWIN.

9. References

[1] A. Hughes, "Here's How Strange 2050 Will be, According to the World's Leading AI Expert," *BBC Science Focus*, 2024. https://www.sciencefocus.com/future-technology/heres-what-2050-will-look-like-according-to-the-godfather-of-ai

[2] A. R. Malik *et al.*, "Exploring Artificial Intelligence in Academic essay: Higher Education student's Perspective," *International Journal of Educational Research Open*, vol. 5, p. 100296, Dec. 2023, doi: https://doi.org/10.1016/j.ijedro.2023.100296.

[3] P. Novikov and A. Kiseleva, "Over-Reliance on Technology in Foreign Language Learning: Case Study of LLB Undergraduates," in *INTED2024 Proceedings*, 2024, pp. 2962–2966.

[4] M. Kuş, "A meta-analysis of the Impact of Technology Related Factors on Students' Academic Performance," *Frontiers in Psychology*, vol. 16, Feb. 2025, doi: https://doi.org/10.3389/fpsyg.2025.1524645.

[5] C. Zhai, S. Wibowo, and L. D. Li, "The Effects of over-reliance on AI Dialogue Systems on students' Cognitive abilities: a Systematic Review," *Smart Learning Environments*, vol. 11, no. 28, pp. 1–37, Jun. 2024, doi: https://doi.org/10.1186/s40561-024-00316-7.

[6] S. Zahidi, "WEF Future of Jobs Report 2023," May 2023. Available: https://www3.weforum.org/docs/WEF_Future_of_Jobs_2023.pdf

[7] H. Haste and V. Chopra, *The Futures of Education for Participation in 2050: Educating for Managing Uncertainty and Ambiguity*. Paris, France: United Nations Educational, Scientific, and Cultural Organization, 2020. Available: https://unesdoc.unesco.org/ark:/48223/pf0000374441

[8] UNESCO, "Artificial Intelligence in education: UNESCO Advances Key Competencies for Teachers and Learners," *Unesco.org*, 2025. https://www.unesco.org/en/articles/artificial-intelligence-education-unesco-advances-key-competencies-teachers-and-learners

[9] S. Weale, "UK Universities Warned to 'stress-test' Assessments as 92% of Students Use AI," *The Guardian*, Feb. 26, 2025. https://www.theguardian.com/education/2025/feb/26/uk-universities-warned-to-stress-test-assessments-as-92-of-students-use-ai

[10] OECD, "Building the Future of Education," Organisation for Economic Cooperation and Development, 2022. Available: https://www.oecd.org/content/dam/oecd/en/about/directorates/EDU/618066-future-of-education-brochure.pdf/_jcr_content/renditions/original./618066-future-of-education-brochure.pdf

[11] Z. Saeed, F. Mancini, T. Glusac, and P. Izadpanahi, "Future City, Digital Twinning and the Urban Realm: a Systematic Literature Review," *Buildings*, vol. 12, no. 5, p. 685, May 2022, doi: https://doi.org/10.3390/buildings12050685.

[12] J. Wang *et al.*, "Revolutionizing Network Autonomy with Intent-Based Network Digital Twins: Architecture, Challenges, and Innovations," *IEEE Communications Magazine*, vol. 63, no. 8, pp. 85–91, Aug. 2025, doi: https://doi.org/10.1109/mcom.001.2400474.

[13] E. Stern and H. Richardson, "Behavioural Modelling of Road users: Current Research and Future Needs," *Transport Reviews*, vol. 25, no. 2, pp. 159–180, Mar. 2005, doi: https://doi.org/10.1080/0144164042000313638.

[14] A. V. Shvetsov and S. Hamood Alsamhi, "When Holographic Communication Meets Metaverse: Applications, Challenges, and Future Trends," *IEEE Access*, vol. 12, pp. 197488–197515, 2024, doi: https://doi.org/10.1109/access.2024.3514576.

[15] Z. Huang and L. Cao, "Quantitative Phase Imaging Based on holography: Trends and New Perspectives," *Light Science & Applications*, vol. 13, no. 1, Jun. 2024, doi:

https://doi.org/10.1038/s41377-024-01453-x.

[16] M. Moller and C. Vuik, “On the Impact of Quantum Computing Technology on Future Developments in high-performance Scientific Computing,” *Ethics and Information Technology*, vol. 19, no. 4, pp. 253–269, Aug. 2017, doi: https://doi.org/10.1007/s10676-017-9438-0.

[17] OECD, “OECD Future of Education and Skills 2030: OECD Learning Compass 2030,” 2019. Available: https://www.oecd.org/content/dam/oecd/en/about/projects/edu/education-2040/1-1-learning-compass/OECD_Learning_Compass_2030_Concept_Note_Series.pdf

[18] F. Miao and W. Holmes, *Guidance for Generative AI in Education and Research*. United Nations Educational, Scientific and Cultural Organization, 2023. doi: https://doi.org/10.54675/ewzm9535.

[19] S. Kanthasamy and S. K. Kingsmill, “Privacy Risk Study 2023,” International Association of Privacy Professionals, Jun. 2023. Available: https://iapp.org/media/pdf/resource_center/privacy_risk_study_2023_executive_summary.pdf

[20] Z. Zakaria, “Digital Blending Learning as a Catalyst for Education Transformation,” KPMG, 2021. Available: https://assets.kpmg.com/content/dam/kpmg/sa/pdf/2021/future-of-learning.pdf

[21] MATSH, “Effectiveness of Personalized Learning: Statistics on Outcomes in Diverse Educational Settings,” *Matsh People Development*, Dec. 28, 2024. https://www.matsh.co/en/statistics-on-personalized-learning-effectiveness/

[22] R. van Hooijdonk, “How VR, AR, and Holograms Are Transforming Education and Training,” *Richardvanhooijdonk.com*, Jan. 06, 2023. https://blog.richardvanhooijdonk.com/en/how-vr-ar-and-holograms-are-transforming-education-and-training/